# SMART LIBRARY CIRCULATION AND RESERVATION SYSTEM

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## 1. Problem Statement

Many libraries experience difficulties managing book borrowing, returns, reservations, and overdue fines manually. This can lead to inaccurate records, poor tracking of book availability, and delays in library services.

## 2. Proposed Solution

This project proposes an Oracle Database 21c and PL/SQL system to manage books, members, loans, returns, reservations, and fines. It will track book availability, calculate overdue fines, and generate reports.

## 3. Data Source

The project will use the University of Illinois at Urbana-Champaign Library Interlibrary Loan Lending Data (2009–2013).

Link: https://databank.illinois.edu/datasets/IDB-6701059

The downloaded Excel workbook contains 169,890 transaction records and 51 columns, including lending dates, titles, authors, ISBNs, formats, and borrowing institution information.

Limitation: The dataset covers institutional interlibrary lending, not individual member loans or reservations. Clearly identified sample records will supplement these missing functions.

## 4. Planned Database Tables

BOOKS, AUTHORS, CATEGORIES, BOOK_COPIES, MEMBERS, LOANS, RESERVATIONS, FINES, PAYMENTS, and INTERLIBRARY_LOANS.

## 5. Planned PL/SQL Features

- Procedures: Register members, issue and return books.

- Functions: Calculate fines and check availability.

- Cursors: Generate loan and overdue reports.

- Packages: Organize library operations.

- Triggers: Update copy availability and record activities.

- Exception handling: Manage invalid transactions and unavailable books.

## 6. Innovation

An interactive Oracle APEX dashboard connected to Oracle Database 21c will allow authorized users to manage library activities. Core business rules will be implemented in PL/SQL.

## 7. References

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